PHYSICAL CHEMISTRY CUMULATIVE EXAM

February 27, 1999

Photons, just like heavy particles, have mechanical momentum, given by h/λ where h is Planck's constant and λ is the wavelength. At Earth's distance from the Sun, the intensity of sunlight is about 1350 W/m^2 outside the atmosphere. The dominant wavelength is about 500 nm (green). To simplify the calculations, assume in answering (a) and (b) that *all* photons emitted by the Sun have a wavelength of 500 nm. During calculations carry the dimensions (units) at all time.

(a) What is the photon flux in $cm^{-2}s^{-1}$, i.e. the number of photons which hit a 1 cm^2 area perpendicular to sunlight? (Note: be careful with m^2 and cm^2)

(b) What is the force arising from light pressure on a 1 m^2 absorbing surface of a spacecraft, perpendicular to the Sun's rays? (Note: the force is equal to the rate of the change of momentum, i.e. momentum/time.)

(c) According to Planck, the energy radiated by a black body at absolute temperature T in a narrow wavelength range $d\lambda$ is proportional to $r(\lambda)=\lambda^{-5}[\exp(hc/\lambda kT]-1)^{-1}d\lambda$ where $\exp(x)=e^x$, and k is the Boltzmann constant, and *c* is the speed of light. The maximum of $r(\lambda)$ as a function of λ is, in very good approximation, at $\lambda_{max}=hc/(5kT)$. The energy density of sunlight, a curve proportional to $r(\lambda)$, is given below. Assuming that the Sun is a perfect (black-body) radiator, which is largely true, estimate the temperature of the Sun's surface.

d) Derive the formula for the maximum of λ. Use the well-known identities for differentiation:

$$dx^n/dx = nx^{n-1}$$
$$df(g(x))/dx = (df/dg)(dg/dx)$$

and assume that $\exp(hc/\lambda kT) >> 1$.

Constants:

Planck's $h = 6.62608\times10^{-34}$ Js

Boltzmann's $k = 1.38066\times10^{-23}$ J/K

Speed of light $c = 2.9979\times10^{8}$ m/s